

EE 210L: Digital Circuits and Logic Design Lab

Lab 3: Multiplexers and Demultiplexers

Name: _____

Date: _____

Partner(s): _____

Section: _____

Record measured voltages, not expected ones.

Part 5.1: Basic Multiplexer Operation

Data inputs wired: D0, D2, D4, D6 = +5 V D1, D3, D5, D7 = GND

S2	S1	S0	Selected input	V at Y (V)	Y	V at W (V)	W
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

Is W always the complement of Y?

Part 5.2: The Enable Input

Select code used: _____

Y with $\bar{E} = 0$: _____ V Y with $\bar{E} = 1$: _____ V

What does the enable input do, and why would a shared data bus need one?

Part 5.3: A 3-Variable Function on the MUX

$$F(A,B,C) = A'BC + AB'C' + AB$$

Fill in which data inputs you tied HIGH, then record measurements.

A (S2)	B (S1)	C (S0)	D _n tied to	Measured V _{out} (V)	Logic Output	Expected F
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

How many logic gates did this function require?

Part 5.4: A 4-Variable Function on the MUX

$$G(A,B,C,D) = \Sigma m(0, 2, 5, 7, 8, 10, 13, 15)$$

Residue Table

S2 S1 S0 (A B C)	Rows covered (minterms)	F when D = 0	F when D = 1	Wire D _n to (0, 1, D, or D')
000				
001				
010				
011				
100				
101				
110				
111				

Measured Results

A	B	C	D	Measured V_{out} (V)	Logic Output	Expected G
0	0	0	0			
0	0	0	1			
0	0	1	0			
0	0	1	1			
0	1	0	0			
0	1	0	1			
0	1	1	0			
0	1	1	1			
1	0	0	0			
1	0	0	1			
1	0	1	0			
1	0	1	1			
1	1	0	0			
1	1	0	1			
1	1	1	0			
1	1	1	1			

Part 5.5: Decoder as a Demultiplexer

Record the voltage on every output for each case. Remember the 74LS138 outputs are active LOW.

Address = 010

Data in (pin 6)	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
0								
1								

Address = 101

Data in (pin 6)	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
0								
1								

Which output followed the data input in each case, and what did the other seven do?

Discussion and Conclusions

Address at least: why implementing a function on a MUX needs no minimization, what the residue method buys you, and how a decoder becomes a demultiplexer.
